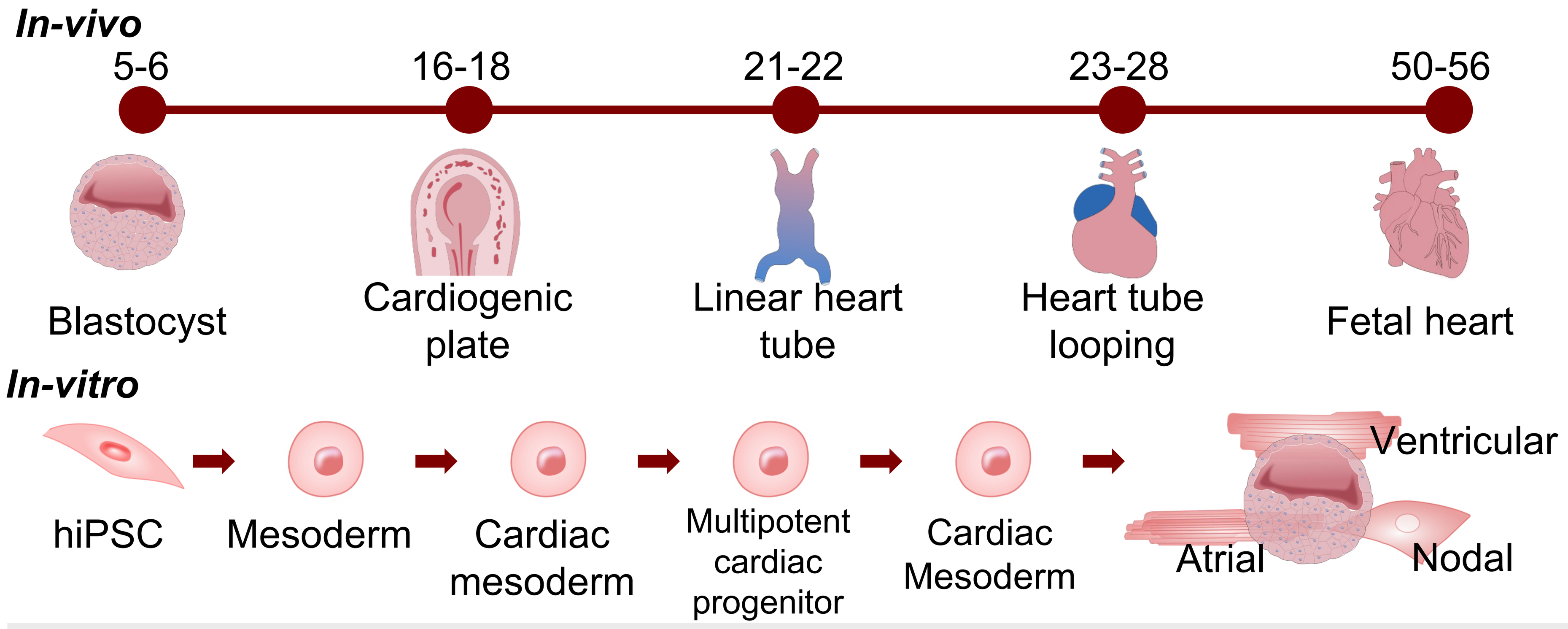
Chamber-Specific Cardiac Organoid Platform with Micropillar-Assisted Calcium and Optical Signal Analysis

Young Hyun Kim[1], Keel Yong Lee[1,2]*
[1] Department of Integrative Bioscience and Biotechnology, Sejong University, Seoul, Korea
[2]Institute of Bioscience and Biotechnology, Sejong University, Seoul, Republic of Korea

ABSTRACT

The cardiac organoid platform integrates micropillar-assisted culture to achieve consistent organoid placement, improved optical assess, and minimized motion artifacts during imaging. Combining calcium imaging and intrinsic optical signal analysis, the system captures lineage-dependent differences in calcium handling and electrophysiological organization. Importantly, the micropillar-assisted configuration enhances experimental reproducibility and **enables high-throughput functional screening** by allowing multiple organoids to be analyzed in parallel with standardize geometry and imaging conditions.

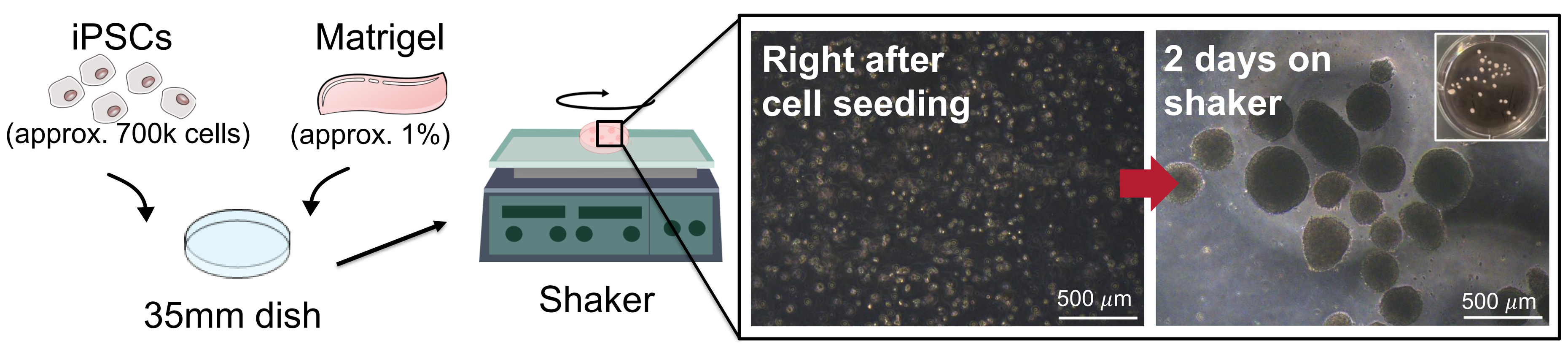
BACKGROUND



Cardiac organoids are widely used to study human cardiogenesis, making reliable structural and functional evaluation essential despite the challenges posed by their heterogenous morphology and spontaneous motion

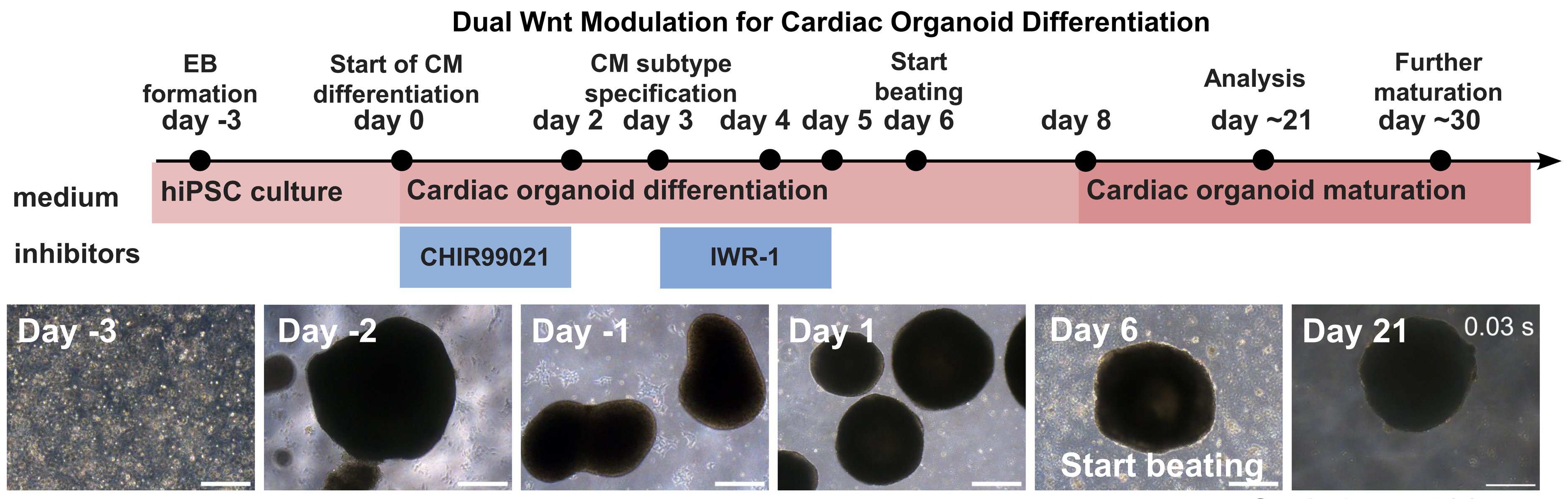
METHOD

Generation of Cardiac Organoid with Shaker



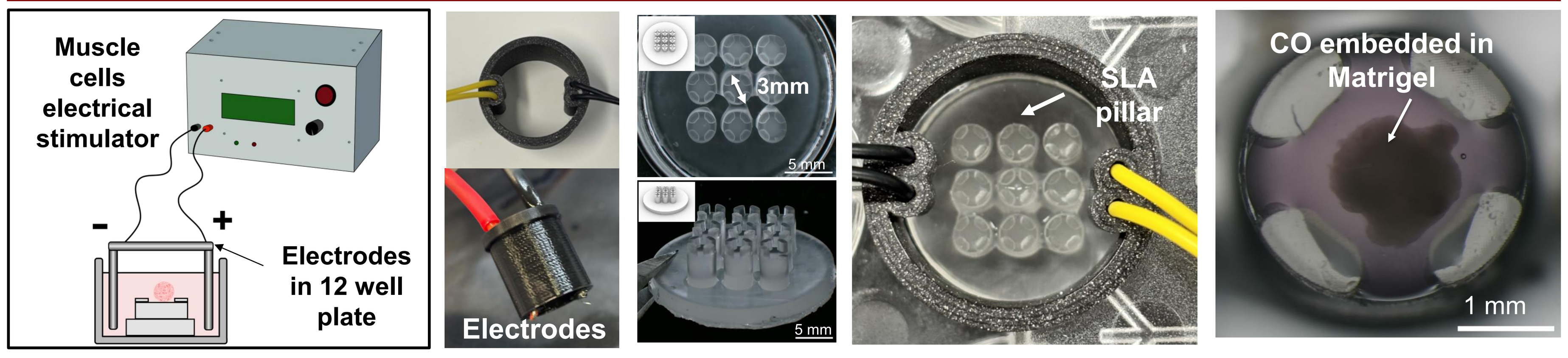
iPSCs were mixed with ECM and cultured on a shaker to form embryoid bodies (Condition: 70rpm on shaker at 37℃ during the entire process))

Differentiation of Embryoid Bodies into Cardiac Organoid



Scale bars: 500 μm

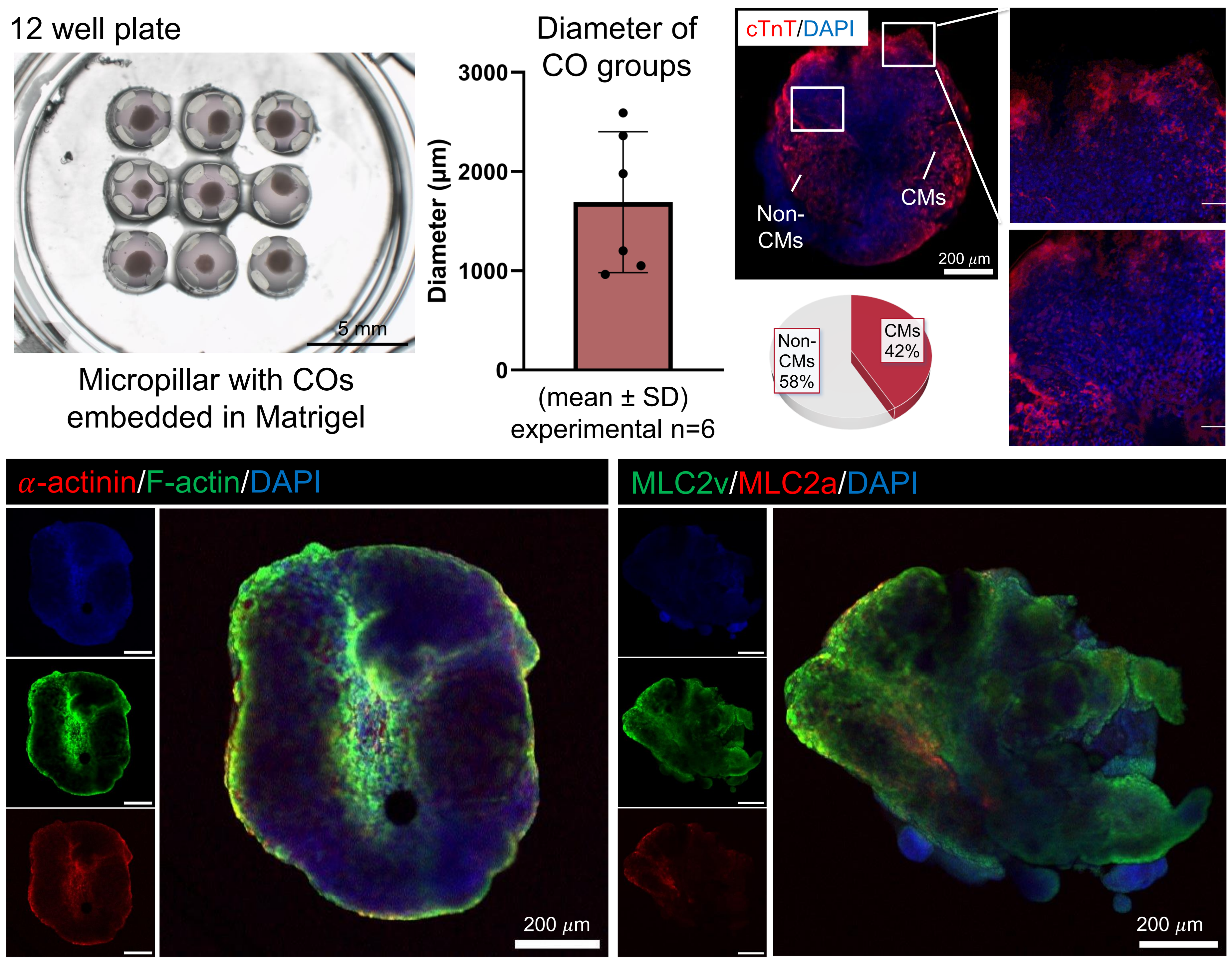
Micropillar-assisted Calcium and Optical Signal Analysis System



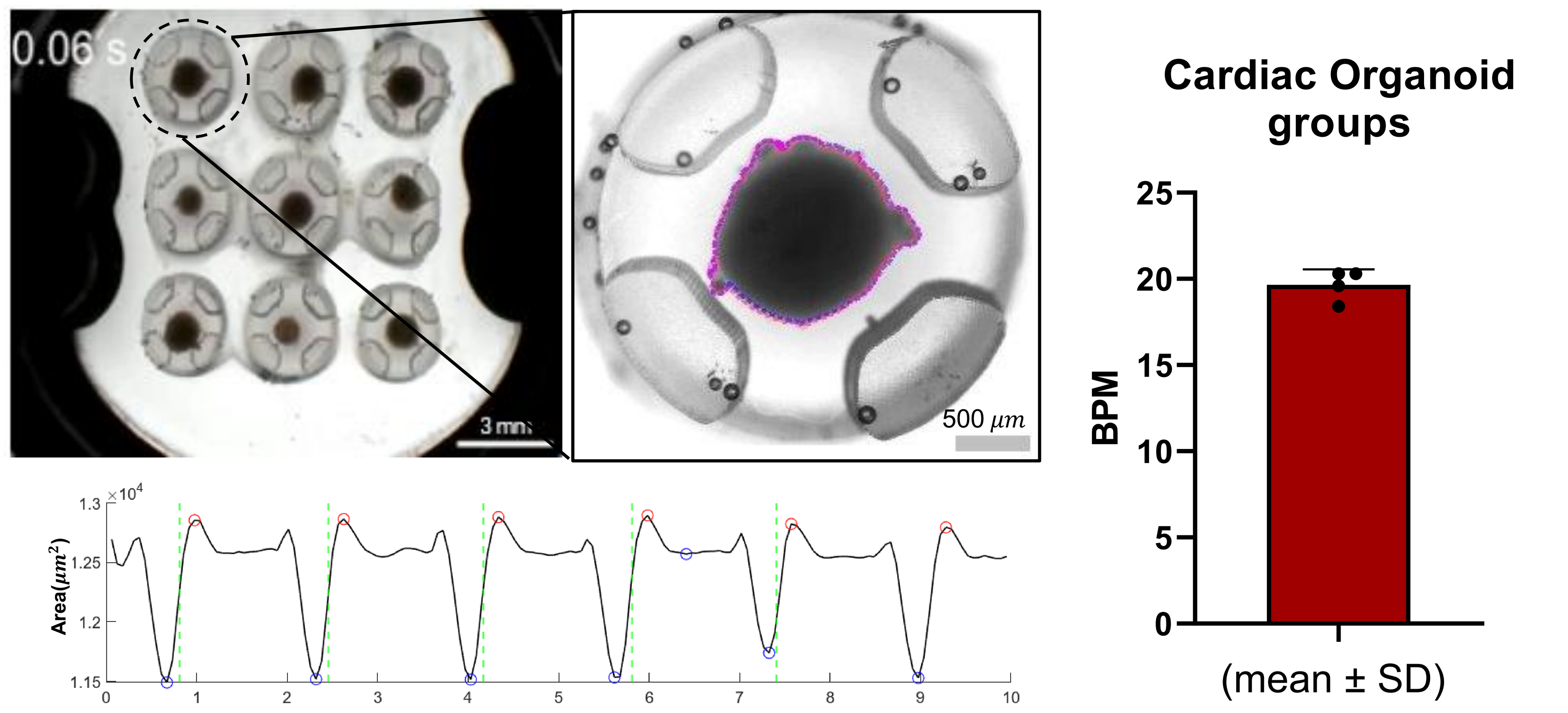
SLA-printed micropillar array designed to fit a 12-well plate and a PLA-based electrode setup used for calcium and intrinsic optical signal measurements

RESULTS

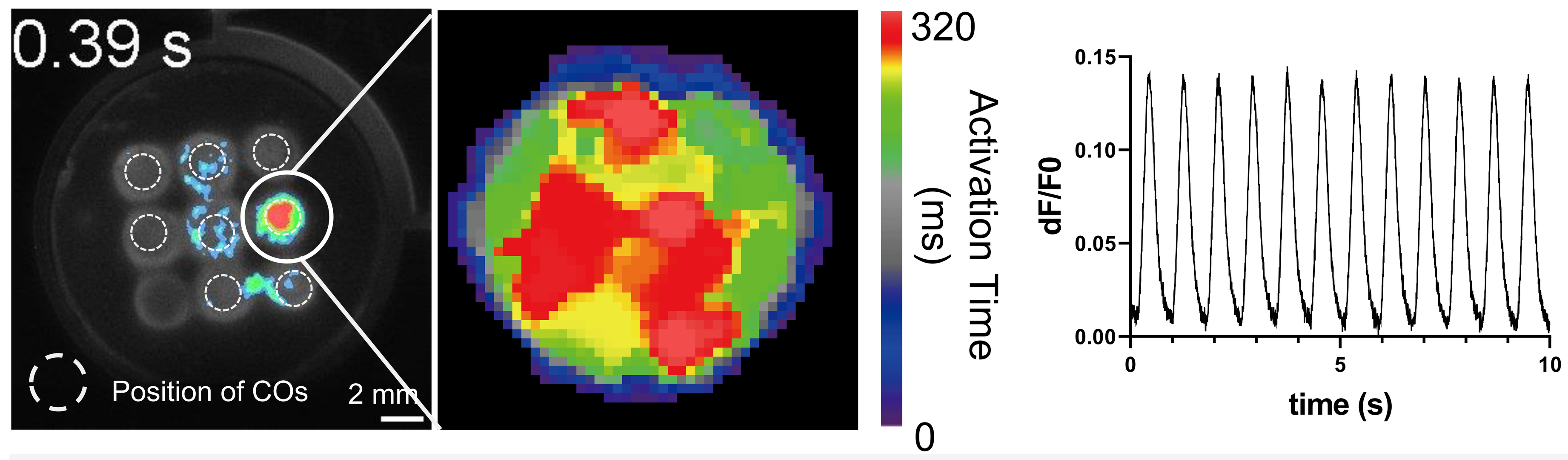
Structural assessment of Cardiac organoid



Functional assessment of Cardiac organoid



Intrinsic optical signal-based mechanical analysis captures both global organoid area changes, offering an integrated view of whole tissue-and regional behavior.



Electrophysiological analysis of cardiac organoids. By measuring of Ca2+ signaling, the distance and time between peaks, conduction velocity cam be determined

Conclusion & Further Study

- **Micropillar-Assisted Platform:** Established a standardized platform that ensures precise organoid positioning and enables high-fidelity, non-invasive dual-signal (optical/calcium) analysis
- **High Reliability & Reproducibility:** Minimized motion artifacts through the micropillar system, providing a robust environment for consistent long-term functional and electrophysiological screening
- **High-Throughput Screening:** Integration of the platform into automated systems for large-scale drug toxicity testing and pharmacological validation
- **Advanced Disease Modeling:** Leveraging chamber-specific organoids to develop precise *in-vitro* models for personalized medicine and complex cardiac pathologies

ACKNOWLEDGMENTS

This research was supported by a grant of the Korea Health Technology R&D Project through the Korea Health Industry Development Institute (KHIDI), funded by the Ministry of Health & Welfare, Republic of Korea (No. RS-2025-24536077), and by a grant from the National Research Foundation of Korea (NRF) funded by the Korea government (MSIT) (No. RS-2025-23323900)

REFERENCE

[1] Lee, Seul-Gi, et al. "Generation of human iPSCs derived heart organoids structurally and functionally similar to heart." *Biomaterials* 290 (2022): 121860.
[2] Kleinsorge, Mandy, and Lukas Cyganek. "Subtype-directed differentiation of human iPSCs into atrial and ventricular cardiomyocytes." *STAR protocols* 1.1 (2020): 100026.